

Definition of a Stochastic Charging model of EV Aggregation and Estimation of its Flexibility Amount

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Abstract—The massive diffusion of Electric Vehicles in the next years is one of the expected outcomes of the Energy Transition. However, this scenario will impact on Medium and Low Voltage electric distribution systems, due to the installation of a multitude of charging infrastructures. Coupled with the accommodation of a plethora of small generation plants (largely photovoltaics), it may cause huge network investments. On the other hand, if smartly managed, they can become a valuable resource of flexibility for the correct electric system operation. Indeed, Electric Vehicles can be considered as mobile storage units that, coordinated by an Aggregator, provide network support by modulating their slow charging at home or workplace. The knowledge beforehand of this resource behavior (expected combined charging profile, available flexibility to adapt this profile and related uncertainties) is essential for Distribution System Operators to be confident in its use and tailor a sustainable network development plan. In the paper, such models have been obtained on the base of Monte Carlo simulations, for different habits of Electric Vehicles drivers.

Keywords—Monte Carlo simulation, Electric Vehicle, Probabilistic modelling, Charging profile

I. INTRODUCTION

The energy transition is a pathway for the next decades toward conversion of the global energy sector from fossil-based to zero-carbon, so reducing energy-related CO₂ emissions to limit climate change. Transport has the highest resilience on fossil fuels of any energy sector and accounts for more than one-third of CO₂ emissions from end-use sectors. Despite the mobility drop in 2020 due to Covid-19 crisis and consequent lockdown measures, transport demand rebounded in 2021 achieving almost the 2019 levels, and passenger and cargo transport is expected to continue increasing rapidly [1]. Therefore, the goal of transport sector emission reduction required by the “Net Zero Emission by 2050” scenario (20% by 2030) needs strong policies to accelerate the shift to the least carbo-intensive travel options. While aviation, shipping, and heavy-duty road freight are harder to decarbonise, Electric Vehicles (EVs) have been growing rapidly around the world, achieving 16 million of cars at the end of 2021 [2]. And this growth is expected to rise in the next decade.

The incorporation of many EV charging facilities, summed to the rise of non-programmable renewable resources, will create significant challenges, particularly at the distribution level. The traditional strategy used by Distribution System Operators (DSOs) to face grid expansion problems is based on reinforcing or building new lines. However, this “Fit & Forget” approach is no longer sustainable in the current electric distribution system scenario, and DSO needs to adopt

one that makes the most of existing assets. The flexibility exploitation from Distributed Energy Resources (DERs) is the most promising and, for this reason, strongly promoted by international scientific organisations [3], regulators [4], and the EU directive 944/2019. Even without considering V2G application, EVs can be considered as flexibility resources because, by anticipating or postponing their charging phase, they can modify their expected demand profile, so providing services to the DSO.

The concept of flexibility is not new in the Literature. There are several studies that analyse how flexibility can be applied to the power system, especially at distribution level. In [5], the authors consider the flexibility service contribution from different sources, such as consumers, producers, and their combination (prosumers), but disregarding contribution from EVs. A flexibility evaluation model has been proposed in [6] to minimize the power imbalance risk from the EV aggregator point of view. In other papers, particular attention has been paid to the use of time series of energy consumption and production. For instance, in [7] a long-term optimisation model is used to evaluate the massive electrification of vehicles on the power system expansion planning and operation. However, the presence of so many elements in the network introduces a certain degree of uncertainty that cannot be captured by deterministic models. For this reason, many authors in the literature introduce probabilistic models and methods as alternatives to classical approaches [8],[9]. In general, the benefits’ maximisation related to the exploitation of flexibility from resources that are not under the direct control of system operators is not easy and requires suitable stochastic models and risk-based network planning tools. In the models’ definition, the main obstacle is the lack of data. Indeed, there is no experience in the behaviour of these new resources and in the flexibility they can provide, and only some pilot projects are in progress. This is particularly true for EVs, where the number of charging facilities is still small.

In the paper, a Monte Carlo Simulation (MCS) has been proposed to reproduce the habits of different EV owners and derive the global electricity demand profile of several slow charging points under the same MV/LV substation. Clearly, this kind of resource is strongly uncertain because the demand may change both in time (when the charge is done) and space (where the EV is connected to a charging facility). The adoption of the MCS allows representing this uncertainty and estimate it in the resulting demand profile. The hypothesised scenario assumes that an Aggregator can manage the charging period of a fleet of EVs and, performing this kind of probabilistic evaluation, provide to DSO the expected day-ahead electricity demand profile (hourly mean value and

standard deviation) and the available flexibility (through the postponement of the EVs' charging).

II. ASSESSMENT OF EVs CHARGING PROFILE

An EV has the capacity to be a flexibility resource due to the potential management of its charging process and its internal battery storage. Suitably aggregated, EVs can be qualified to provide ancillary services, both in up (a decrease of the demand equivalent to a generation increment) and down (an increment of the demand equivalent to a generation curtailment). And it is not necessary to assume V2G capability to have this bi-directional behaviour, because the EV's charging postponement corresponds to a cut of the electric demand, while bringing it forward is equivalent to an increment of the electric demand. However, this last option presumes that the EV's owner links his vehicle to the charger even without charging (motivated by a sort of remuneration from the Aggregator).

The quantification of this flexibility entails the knowledge of the expected charging demand profile of the EVs aggregation, to understand both its impact on the electric distribution network and the room for altering it. This knowledge is a critical point, due to the lack of historical statistics and the uncertain behaviour of EV drivers. Indeed, apart from some systematic routines (e.g., go to work or bring the sons at school), several behaviours may change daily (where and when go shopping, when go to a restaurant, where and when have leisure activities). Moreover, within the same family, it may be different the usage of the main car (to go working) and the second one (to go shopping or leisure and stay parked at home for most of the day). Finally, even within a systematic routine, it may be different the public slow charger used, dependent on the available parking area.

Consequently, an Aggregator that manages several slow charging points, under different Secondary Substations (SSs), may have in the same hour of different days, a different number of EVs connected, with different initial State of Charge (SoC) values, and different plans for the rest of the day. To face with all these uncertainties and build a representative probabilistic model of the EVs aggregation demand in each SS, a similar procedure as that described in [10] has been developed. A MCS approach has been adopted by simulating hundreds of times the behaviour of the same group of EV's owners within a specific area (e.g., an entire city or just some neighbourhoods), determining when the vehicle is connected to a slow charger, where it is connected (at home, at work, in a shopping centre), how long it remains connected and the charging profile along these periods (need/no-need to recharge, minimum SoC requested when the driver moves again from the parking slot). Some drivers live outside of the area under examination, so their EVs do not appear during the night hours (because connected at home under a SS not included in the area). Some others work out of this area (or moves out of it for different reasons) and their EVs may not appears during the daytime. The flowchart of the MCS is depicted in Fig. 1.

The procedure starts with the definition of the mobility scenario. Three aspects must be outlined: the characteristic of the area associated to the Aggregator, the habits of the EV's drivers and the features of the EVs fleet. The main data are indicated in TABLE I. The first category of information defines the boundaries of the test case analysed. Considering slow chargers and typical urban mobility, the area may

represent an entire city or a portion of it (some districts of neighbourhoods), with a given number of SSs, each of them supplying residential loads, offices, small shops, large shopping centres, parks, sport facilities, etc. Under each SS, several slow and fast chargers are available, located in residential buildings (wall boxes), at work, in public parking areas, etc. This number has been fixed for each SS at the beginning of the study.

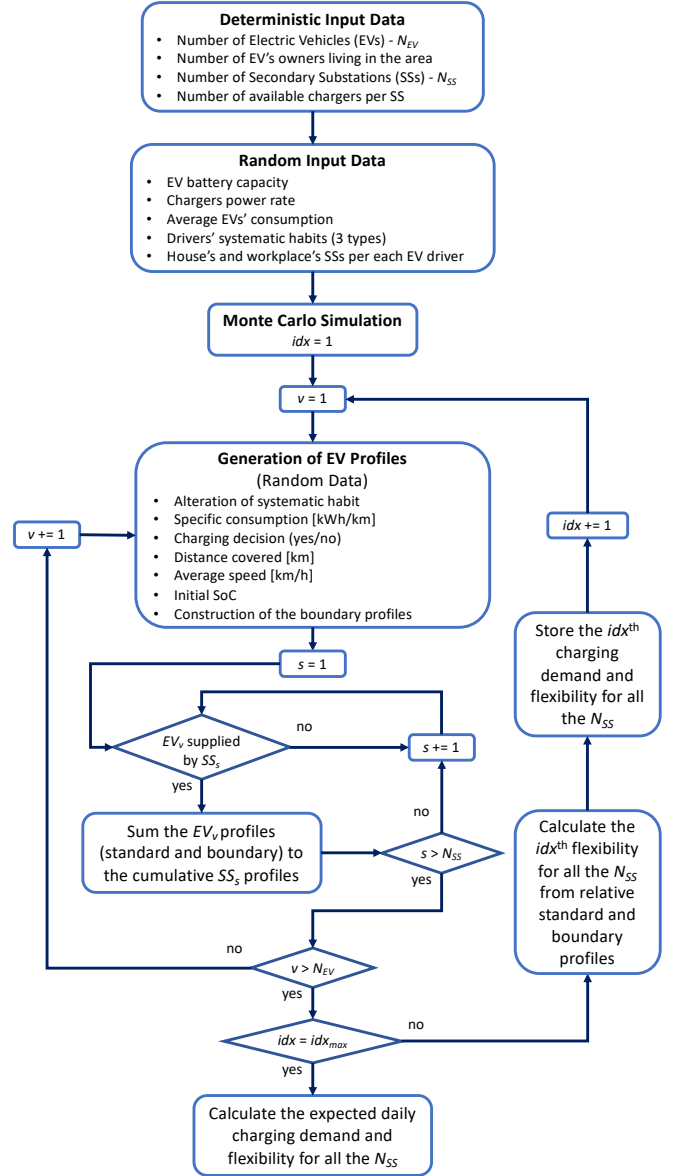


Fig. 1. Monte Carlo Simulation flowchart used to emulate EVs mobility and charging demand.

TABLE I. MAIN DATA USED FOR MOBILITY SCENARIO DEFINITION

Area characteristic	Driver's habits	EVs features
Number of EVs assumed in the study	Type of systematic habit	EV's battery capacity
Number of EV's owners living in the area	Supplying SSs for house and workplace per EV's driver	Average consumption
Number of Secondary Substations (SS)		Slow charger power rate
Number of available slow chargers per SS		

Another category of data to be defined is the EVs' and chargers' technical features. In addition to the number of EV chargers under each SS, it is essential to fix the power rates of these chargers. Disregarding rapid and ultra-rapid charging facilities (with rate greater than 50 kW DC and 43 kW AC), different rates are available for residential and public chargers (at home, at work, along streets), from 3 kW up to 22 kW. In the study, it has been considered only chargers of 3 kW (85%), 7 kW (10%) and 10 kW (5%). Regarding EV's features that influence the study, it is necessary to fix the EV battery capacity and the EV average consumption. The first ranges from a minimum of 20 kWh of the small city cars up to 100 kWh and more for large SUV and pickup. In the study an average value of 57 kWh has been assumed and the battery capacity has been fixed by extracting randomly its value with a normal distribution. The average consumption has been randomly assigned to each EV with an average of 0.2 kWh/km.

The last category that defines the study concerns the driver's habits, that means his systematic routines, his average daily driven distance. The driver behaviour has been defined by giving the hours of the typical day when the EV is connected to a charger facility or when it is in motion. Three behaviours have been fixed, as represented in Fig. 2. The first two refer to full-time workers with 40 hours' work per week (8 hours per working day). The difference between them is in the lunch break: during this period (from 13:00 to 14:00) the first type of drivers leaves the EV connected to the charger (forming a continuous interval of 9 hours) and has the lunch near the working place, while the second type uses the EV to reach the location for lunch so creating two separated intervals of 4 hours of EV connected to the charger. The third behaviour has been designed more variable, simulating a family second vehicle. These habits are assigned randomly to the EV drivers of the study with the following share: 50% of EV drivers follows the first habit, 30% the second one and 20% the third one. A second important feature to define in this category is the daily covered distance. For each driver of the first two habits is assigned the SS that supplies his house (with a private wall box charger) and the SS closest to his workplace. By so doing, the driven distance can be estimated, together with the EV's consumption.

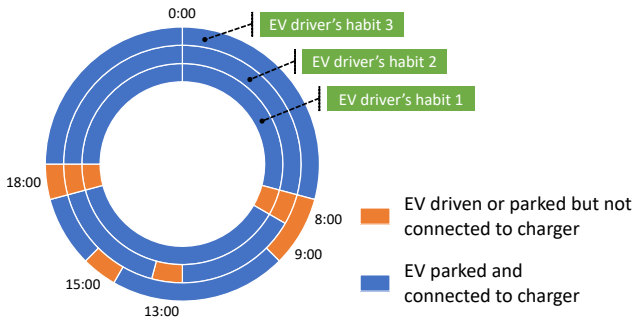


Fig. 2. Systematic routines of EV's drivers.

After all these assignments, the driving scenario is defined and the MCS procedure can start. A typical day is repeated several times and, by extracting randomly (from normal or uniform distributions) all the uncertain data, the exact behaviour of each EV's driver is created. Among this uncertain data there are the minute of departure from home, the distance covered (that depends also on the presence of available public chargers at the destination), the specific

consumption during the trip, the average speed, and the initial SoC of the EV's battery at home (from a minimum level accepted by the driver up to the full battery capacity). For each Monte Carlo iteration, the EV driver's habits can be modified to include occasional events like advancing or postponing the check out or dining out.

The EV movements are defined starting from the two SSs that supply the initial and final points of the trip (known for drivers' habits of type 1 or 2, randomly selected for driver's habit of type 3). The two SSs can also be the same. By knowing the size of the area supplied by each SS and the relative distance between the departure and arrival SSs, the covered distance is randomly estimated around the calculated average distance. A specific consumption is also extracted randomly around the average value assigned to each EV to simulate different traffic conditions. By so doing, the SoC decrement can be approximated with the following linear equation:

$$SoC_{arrival}^i = SoC_{departure}^i - D_i \cdot C_i \cdot \frac{100}{BC_i} \quad (1)$$

where $SoC_{arrival}^i$ and the $SoC_{departure}^i$ are the states of charge of the i^{th} EV at the beginning and the end of the trip, D_i is the distance covered in the specific trip (km) by the i^{th} EV, C_i is the specific consumption of the i^{th} EV (kWh/km), and BC_i is total battery capacity of the i^{th} EV (kWh).

Regarding the average speed (v_i), this value has been estimated assuming a linear inverse dependence on the consumption. Indeed, the "green speed" of an electric vehicle (i.e., the one with the minimum consumption) is in the range 60–70 km/h, while if the vehicle drives above or below the green speed, it consumes more [11]. As an example, the maximum energy consumption for zero acceleration is 0.25 kWh/km at 0 km/h, and decreases to the minimum of 0.1 kWh/km at 60 km/h. The correlation is more complex and depends also on acceleration and the characteristics of the EV. Assuming that for urban movements difficultly the average speed is higher than this green speed, the simplified assumption made in the paper is acceptable. Thus, when the specific consumption is randomly extracted, the average speed is immediately assigned. This value is used to assess the minute of arrival as:

$$t_{arrival}^i = t_{departure}^i + 60 \cdot \frac{D_i}{v_i} \quad (2)$$

Considering the hour discretization used to represent the results, the knowledge of the minute of departure and arrival is used to identify the hours of the day into which the EV is parked and connected to a charger.

For the exact definition of the charging demand when the EV is connected to a charger, two SoC levels are randomly extracted: the minimum SoC requested at the end of the parking interval and a minimum threshold at the beginning of the parking interval to activate the possible driver's choice to recharge or not his EV (leaving the vehicle connected to the charger). This choice has been simulated as the toss of a coin with a higher weight for charging. When the EV is connected to the charger and the driver chooses to recharge, the energy absorption starts immediately at the nominal charger's power rate and ends when the battery has achieved the minimum requested SoC. This behaviour is used to build the standard charging demand profile.

Two boundary profiles are used to approximately estimate the available flexibility: the first is obtained by postponing as much as possible the recharge of the EVs, the second is derived by activating immediately the recharge of the EVs connected in stand-by (drivers' choice). By comparing the standard profile with these boundaries, a rough estimation of the amount of flexibility (increment or decrement of the demand in each hour of the typical day) can be obtained.

All these profiles (standard and boundary) are calculated in each iteration of the MCS. So, at the end of the procedure, the following outputs are provided for each secondary substation of a given electric distribution system:

- the expected daily consumption profiles of the aggregation of charging points,
- the uncertainty around these profiles,
- the available aggregated up and down power for DSO services (the flexibility),
- hour by hour, the uncertain range of this flexibility.

The first two data can be used within the probabilistic load flow of a modern planning tool to determine the impact of the fleet of EVs on the supplying network. The last two, instead, are useful to check if the available flexibility can contribute to solve technical issues.

It has been assumed that the Aggregator does not manage the same prefixed number of vehicles independently from the charging points selected, but the same number of charging point under several MV/LV secondary substations (assumed as interfaces between the Aggregator and the Distributor). Thus, the Aggregator provides to the DSO for the next day the forecasted daily consumption profile of its charging points and the expected amount of flexibility.

III. RESULTS

The described methodology has been tested on a real MV distribution network (Fig. 3). The network has been used just to provide the total number of MV/LV Secondary Substations (22) and to calculate the relative distances. It has been assumed that it feeds a portion of an urban area. The SSs are classified depending on the type of customers supplied: almost all residential users, a mix of residential – commercial – office users, and a large shopping center. Being the study focused on the definition of the EV charging demand, no other additional information is required from the electric distribution network. However, if the results of this research are applied to planning studies, also the demand profile of the existing loads shall be provided and added to the obtained EV charging profiles.

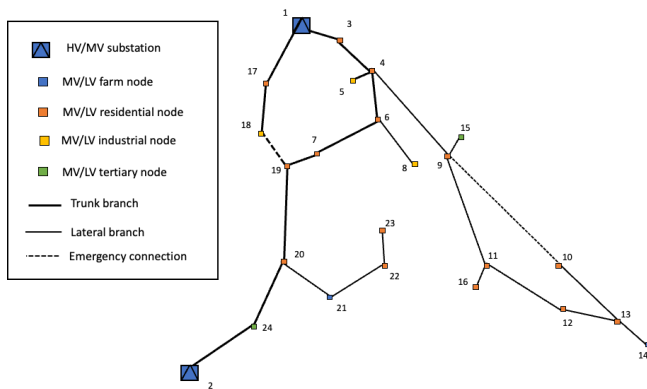


Fig. 3. Tested electric distribution network

A single typical day has been used to model the user behaviors. Obviously, the procedure is easily adaptable to include multiple typical days, differentiated by season, workdays, and weekends. Some attention should be put on the definition of drivers' habits in different days.

The study assumes the presence of 1000 EVs, while the number of available chargers under each SS varies from 50 (in pure residential neighborhoods) to 100 (in city center or large parking areas). The MCS procedure has been executed for 10000 iterations to achieve stable results.

In Fig. 4 the expected daily charging demand profiles for different types of customers are reported. The first one is a typical city center SS, with a mix of residential, commercial, and offices loads. Therefore, during the night the EVs' demand is formed by those citizens that live in the same neighborhood. In European city centers it is common the presence of apartment blocks without private parking. Consequently, the great part of the charging facilities are represented by public chargers along the streets or in public and workplace parking areas. In the early morning, the majority of resident EV's drivers moves out of it to achieve their workplaces, and they are substituted by different EV's drivers that come to work or for shopping purposes and connect their vehicles to the available public chargers, left vacant by the residents. In the evening, the area witnesses the opposite movements of EV's drivers.

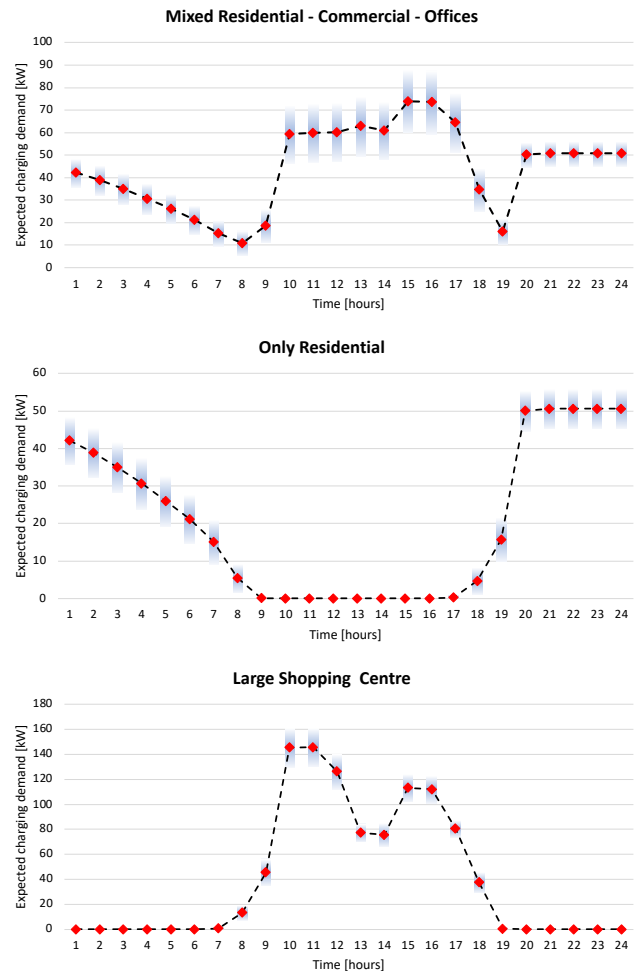


Fig. 4. Expected charging demand daily profiles, for different type of customers supplied by a Secondary Substation

The second example refers to a pure residential neighborhood, characterized by several independent houses or small apartment blocks, with frequent availability of private parking. Therefore, charger facilities are typically formed by private wall-boxes, that are not available to other EV's drivers. Consequently, when the residents go to work the EV charging demand falls almost to zero (except for few EVs used as second cars in a family).

The third case is related to a large shopping center that generally offers to its clients a large parking area. However, this area is typically exploited when the shopping center is open, remaining empty during the night.

In the three profiles of Fig. 4 it has been depicted also the uncertainty band of the expected demand for each hour of the typical day. This information is essential for a correct application of modern probabilistic planning tools, developed to minimize the risk of technical constraints violation under an acceptable limit, rather than nullify it as in the classical "Fit & Forget" planning approach [8],[12].

In Fig. 5 the flexibility calculated from the first case of Fig. 4 has been reported, differentiated between upward (increment of the demand) and downward (cut of the demand) services. The first service has been obtained by activating the battery recharge of those EVs left connected in standby. On the contrary, the second service is obtained by stopping the charging operation. The hypotheses adopted for the MCS bring the EV's batteries to be often at high levels of SoC and, consequently, they can recharge at the minimum SoC level in few hours. This situation allows stopping the recharge with low probability of "payback effect", i.e., without the need to restart the charge after the cut. The result is that the downward flexibility is almost coincident to the expected charging profile. A more sophisticated description of the mobility scenario will probably change this result, together with a more accurate assessment of the available flexibility.

Also for this result, the developed tool provide the hourly expected value and the associated uncertainty. This probabilistic model is useful to estimate the residual risk in the exploitation of flexibility as alternative to traditional planning solutions (like network reinforcements), helping in the development of robust network expansion plans [13].

IV. CONCLUSIONS

The paper illustrates a methodology for estimate the expected charging demand of an aggregation of EVs connected to slow charging facilities, used typically by those drivers that leave their car parked for many hours consecutively. The obtained models are important to understand the impact of a large diffusion of EVs in the electric distribution systems. Additionally, these models are probabilistic, i.e., they provide not only the expected demand but also the relative standard deviations (uncertainties). This characteristic is important for the correct application of the modern electric distribution planning tools, based on risk minimization. The developed tool provides also a rough estimation of the available flexibility that can be exploited from this aggregation.

Future works intend improving the representation of the mobility scenario and refine the flexibility calculation, to provide DSO more reliable probabilistic models that will support the development of the upcoming and even more smart electric distribution systems.

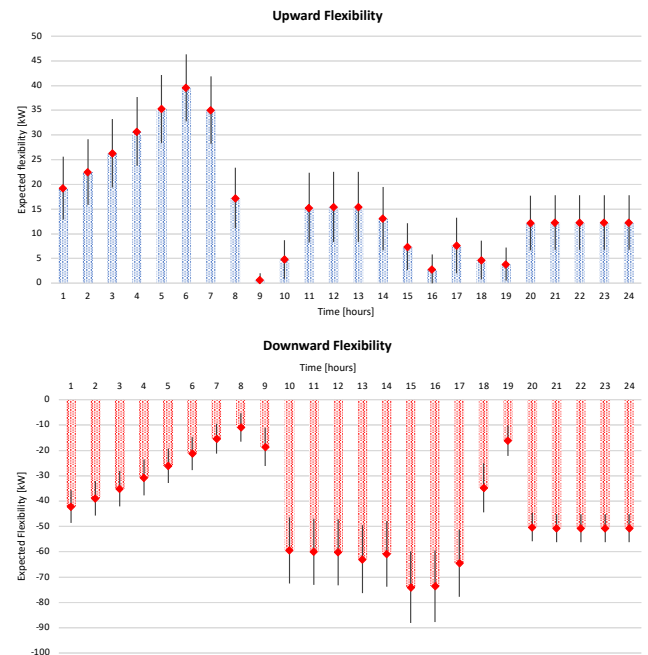


Fig. 5. Flexibility from EVs' aggregation of the first case of Fig. 4 (mix of residential – commercial – offices)

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